

TØ 13 – Mikroøkonomi 2 – 2024

Problem 1

Consider the following labor market. There is one worker and two firms.

The timing of the game is as follows:

- Nature draws the worker's ability (type) $\theta \in \{1, 2\}$. The prior probability of $\theta = 2$ is 0.1.
- The worker learns his type.
- The worker can invest in his education level $e \geq 0$.
- Each firm makes a wage offer (w) to the worker (they make their offers simultaneously). They do not observe the worker's type (the prior probability distribution is however common knowledge), but they observe the education level.
- The worker accepts the higher wage offer. If they are equal, he randomizes 50-50.

Payoffs are as follows. If a firm hires a worker of productivity θ , its payoff is $\Pi(\theta, w) = \theta - w$. The utility of the worker is $u(w, e, 2) = w - 0.4e$ if his type is $\theta = 2$ and $u(w, e, 1) = w - e$ if his type is $\theta = 1$.

Part 1:

1. Argue which equilibrium concept seems relevant in this context and why.
2. What are the requirements for an equilibrium?

Theory derived in this problem is from L10 and L11

1

Dynamic game of incomplete information so PBE

2

Strategy profile σ^* and system of beliefs μ^* satisfying

Requirement 1: At each information set, the player with the move must have a belief about which node in the information set has been reached by the play of the game. For a non-singleton information set, a belief is a probability distribution over the nodes in the information set; for a singleton set, the player's belief puts probability one on the single decision node.

Requirement 2: Given their beliefs, the players' strategies must be sequentially rational. That is, at each information set the action taken by the player with the move (and the player's subsequent strategy) must be optimal given the player's belief at the information set and the other player's subsequent strategy (where a subsequent strategy is a complete plan of action covering every contingency that might arise after the given information set has been reached).

Requirement 3: The players' beliefs in every information set (on and off the equilibrium path) are determined by the players' equilibrium strategies and Bayes' rule whenever applicable.

Part 2: Separating equilibria:

3. Argue that type $\theta = 1$ chooses an education level of 0 in any separating equilibrium.
4. Derive the minimum education level of type $\theta = 2$ ($\hat{e}$) that can be supported in a separating equilibrium.
5. Specify wages and beliefs of the firms (on and off the equilibrium path) that support a separating equilibrium in which type $\theta = 2$ chooses an education level $\hat{e}$ and type $\theta = 1$ chooses an education level of 0.
6. Illustrate this separating equilibrium graphically in (e, w) - space.
7. Derive the maximum education level of type $\theta = 2$ ($\tilde{e}$) that can be supported in a separating equilibrium.
8. Specify wages and beliefs of the firms (on and off the equilibrium path) that support a separating equilibrium in which type $\theta = 2$ chooses an education level $\tilde{e}$ and type $\theta = 1$ chooses an education level of 0.

3

Type $\theta = 1$ is the low type since it is more costly for type 1 to perform effort:

$$u_1 = w - e$$

$$u_2 = w - 0.4e$$

Reason is—very standard argument—that type 1 will never pretend to be type 2 and since education reduces her utility, she will choose the minimum education, i.e., $e = 0$

Note important to the next part: since it is a competitive market, profit of the firm will be zero so:

$$w_L = \theta_L = 1$$

$$w_H = \theta_H = 2$$

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In screening, ICL must hold so

$$w - e = \theta_L$$

Inserting the numbers

$$1 = 2 - e \Leftrightarrow e = 1$$

5

The equilibrium we have derived consists of strategies

$$e^*(\theta) = \begin{cases} 0 & \text{if } \theta = \theta_L \\ \tilde{e} & \text{if } \theta = \theta_H \end{cases}$$

for the worker and

$$w^*(e) = \begin{cases} \theta_H & \text{if } e \geq \tilde{e} \\ \theta_L & \text{if } e < \tilde{e} \end{cases}$$

for the firms, and beliefs

$$\mu(e) = \begin{cases} 1 & \text{if } e \geq \tilde{e} \\ 0 & \text{if } e < \tilde{e} \end{cases}$$

Wages:

$$w(e) = \begin{cases} 2, & e \geq \tilde{e} \\ 1, & e < \tilde{e} \end{cases}$$

If the education is $e = 1$, the firms are sure they are dealing with a high type and hence, they will offer the wage based on high productivity, $w_H = \theta_H = 2$.

When the education level is $e \neq 1$, the firms know that it is a worker of low type, so offer a low wage, $w_L = \theta_L = 1$.

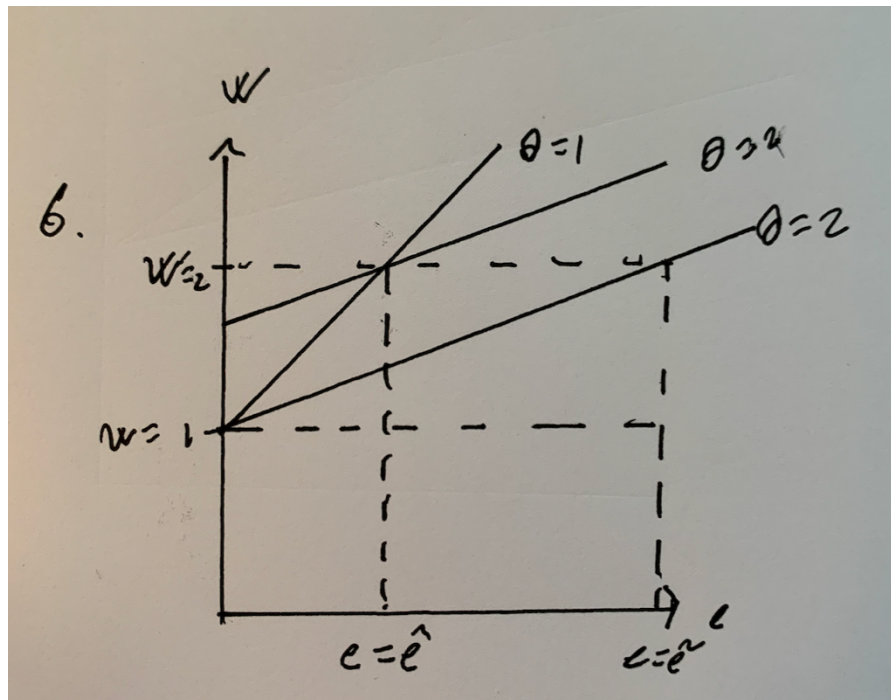
Beliefs are:

$$\mu(e) = \begin{cases} 1, & e \geq \tilde{e} \\ 0, & e < \tilde{e} \end{cases}$$

Explanation:

If the education level is $e = 1$, the firms are sure that it is a worker of high type.
 Conversely, if the education level is $e \neq 1$, the firms are sure that it is a low type.

6



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ICH yields

$$2 - 0.4e \geq 1 \Leftrightarrow e \leq 2.5$$

Maximum education level is $\tilde{e} = 2.5$

8

Beliefs are now

$$\mu(e = 2.5) = 1$$

$$\mu(e \neq 2.5) = 0$$

And wages

$$w(e = 2.5) = 2 = \theta_H$$

$$w(e \neq 2.5) = 1 = \theta_L$$

For explanation, please see part 5

Part 3: Pooling equilibria:

9. Derive the highest education level ($\check{e}$) that can be supported in a pooling equilibrium.
10. Specify wages and beliefs of the firms (on and off the equilibrium path) that support a pooling equilibrium in which the worker chooses an education level $\check{e}$.
11. Illustrate this pooling equilibrium graphically in the (e, w) -space.

Part 4: Refinements:

Marie-Louise will cover this part in the lecture on Friday, May 10

12. Show with an appropriate equilibrium refinement for signaling games how to eliminate all separating equilibria in which type $\theta = 2$ chooses an education level that is strictly larger than the education level $\hat{e}$ derived in part 2.

9

I calculate the wage

$$\mathbb{E}[\theta] = p(\theta = 2) \cdot \theta_H + (1 - p(\theta = 2)) \cdot \theta_L = 0.1 \cdot 2 + 0.9 \cdot 1 = 1.1$$

Which implies that if the low type will choose an education level greater than $e = 0$, the following must be satisfied (since the low type will get 1.1 instead of 1 but will have disutility from the education):

$$1.1 - \check{e} = 1 \Leftrightarrow \check{e} = 0.1$$

Binding since we are asked to find the highest possible level

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The equilibrium wage schedule is

$$w(e) = \begin{cases} E(\theta) & \text{if } e \geq e^* \\ \theta_L & \text{if } e < e^* \end{cases}$$

$$\text{and beliefs are } \mu(e) = \begin{cases} \lambda & \text{if } e \geq e^* \\ \theta_L & \text{if } e < e^* \end{cases}$$

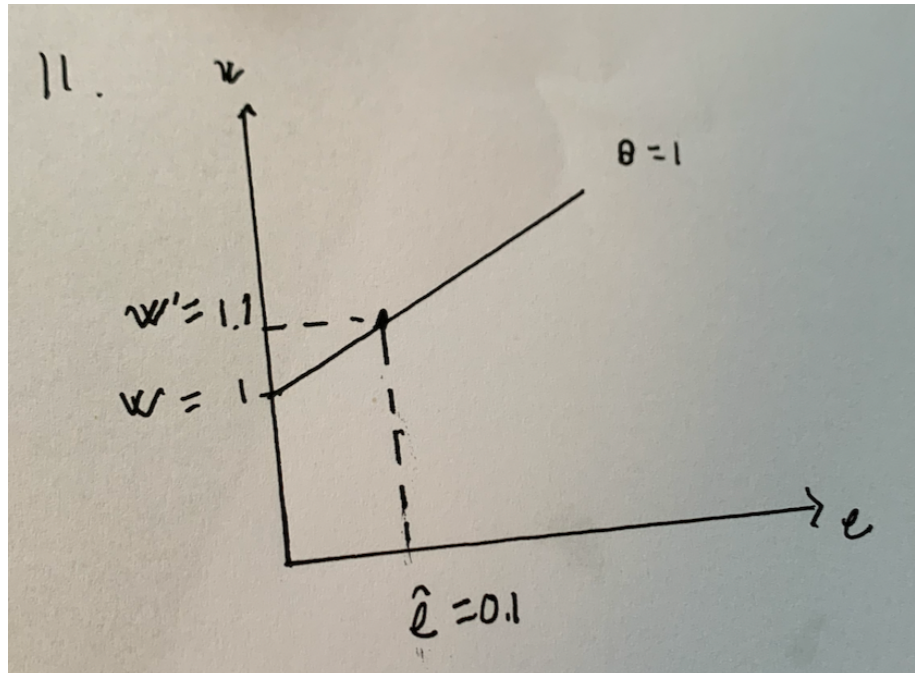
Where λ is $p(\theta = 2) = 0.1$

$$w(e) = \begin{cases} 1.1, & e \geq \check{e} \\ 1, & e < \check{e} \end{cases}$$

Beliefs

$$\mu(e) = \begin{cases} 0.1, & e \geq \check{e} \\ 0, & e < \check{e} \end{cases}$$

11



For problem 2, please see the other PDF

Problem 2

Consider the following signalling game. The current owners of the Firm ABC inc. and an investor are in negotiations. The investor would like to enter as a shareholder of the firm but he is uncertain of whether the firm is in financial distress or not. On the other hand, the owners of the firm know exactly its financial condition. The investor has a prior belief about whether the Firm is in good standing: there is probability $\frac{2}{3}$ that the Firm is in good condition. All of this is common knowledge.

If the firm is in good condition, its profits will be equal to 8 Million, whereas if the firm is in bad condition, its profit will be equal to 4 Million. The investor thinks of entering as a 25% shareholder, and thus having the respective share in the profits. The cost of buying this share is 1 million.

The current owners of the firm are considering ways of signaling the firm's financial situation. They can pay a high or a low dividend. A high dividend will involve returning 75% of the profits to the shareholders, and a low dividend 40% of the profits.

1. Draw the game tree that corresponds to this problem.

Hint: when a firm is in good standing and issues dividends, the current owners receive either $0.75 \cdot 0.75 \cdot 8 + 1 = 5.5$ Million or $0.75 \cdot 8 = 6$ Million in payoffs depending on whether the investor buys equity or not.

2. Describe all the pure-strategy pooling and separating Perfect Bayesian Equilibria.

Problem 3

Consider a monopolist of type Low (L) or High (H). λ is the exogenous probability that the monopolist is type H . The monopolist produces a good of quality i determined by his type ($i \in \{L, H\}$). An H -type monopolist produces at constant marginal costs $c_H = c > 0$ while an L -type produces at constant marginal costs $c_L = 0$. The good is an experience good so the monopolist has private information about the quality of the good. The buyers only learn the quality after purchase.

Consumers are heterogeneous with respect to their taste for quality and thus their willingness to buy. There is a continuum of potential consumers with mass normalized to 1. Denote by θ the parameter which governs the consumers' taste for quality, where $\theta \in [0, 1]$. θ is uniformly distributed.

A consumer of type θ gets utility $u(\theta, i, p) = \theta i - p$ of buying a good of quality i ($i \in \{L, H\}$) at price p . As consumers only learn the quality of the product after purchase they base their decision to purchase or not on the perceived (or believed) quality q^p .

The profit function of the monopolist of type i who is perceived by the consumers as producing with quality q^p (see definition of q^p below) is therefore

$$\Pi_i(q^p, p) = [p - c_i]D(p, q^p)$$

where $D(p, q^p)$ is the aggregate demand (the share of consumers buying at price p from the monopolist who is believed to be of quality q^p).

The timing of the game is as follows:

- Nature determines the type of monopolist.

- The monopolist observes his type and sets price p for his good. The monopolist can only set one price which holds for all consumers regardless of their type θ .
- Consumers observe price p (but not the type of the monopolist) and update their belief $Pr(i = H|p)$. All types of consumers share the same belief.

Given $Pr(i = H|p)$ consumers expect quality

$$q^p = H \cdot Pr(i = H|p) + L \cdot (1 - Pr(i = H|p))$$

- Based on this consumers decide whether to buy the good or not. If the consumer does not buy, his utility is 0.

Below we will refer to the first best as the setting where quality is observed by consumers ex ante (before buying). In the second best setting, the consumers will not observe the quality ex ante. To simplify the solution in the second best we will assume that beliefs off the equilibrium path are "extremely pessimistic" such that any deviation from the equilibrium strategies will imply that the consumer expects the monopolist to be of type L .

1. Determine the aggregate demand for a good of quality q^p i.e. $D(p, q^p)$.
(Hint: note that the marginal consumer $\hat{\theta}$ which buys the good can be determined as: $\hat{\theta}q^p - p = 0 \Rightarrow \hat{\theta} = \frac{p}{q^p}$).
2. State the maximization problem of the monopolist under the first best. Solve for the first best equilibrium: determine price, $D(p, q^p)$ and profit for each type of monopolist.
3. Now consider the second best. Describe the information structure and explain why we should think of this as a signaling game. Identify senders, receivers and messages.
4. Is the first best equilibrium a possible second best separating equilibrium? (Assume that $H(H - L) > c^2$).
5. Is the single crossing property satisfied in this setting? (Hint: how does profits change with price for an L versus H type monopolist and what does that imply for prices as a signal?)
6. Solve for the choice of price p_L^* and resulting profits for a monopolist of low type in a separating equilibrium (second best).

1

$$\tilde{\theta}q - p = 0 \Leftrightarrow \tilde{\theta} = \frac{p}{q}$$

Aggregate demand:

$$\int_{\frac{p}{q}}^1 1d\theta = [\theta]_{\frac{p}{q}}^1 = 1 - \frac{p}{q}$$

2

FB

$$\pi = \max(p - c_i) \left(1 - \frac{p}{q}\right)$$

FOC is then

$$\frac{\partial \pi}{\partial p} = \left(1 - \frac{p}{q}\right) - \left(\frac{p - c_i}{q}\right) = 0 \Leftrightarrow p = \frac{i + c_i}{2}, i \in \{L, H\}$$

And the prices are found by inserting low and high into the FOC:

$$p_{FB}^L = \frac{L + c_L}{2} = \frac{1}{2}L$$

$$p_{FB}^H = \frac{H + c_H}{2} = \frac{H + c}{2}$$

Aggregate demand was

$$D = 1 - \frac{p}{q}$$

So,

$$D(p_{FB}^L, L) = 1 - p_{FB}^L \cdot \frac{1}{q} = 1 - \frac{1}{2}L \cdot \frac{1}{L} = \frac{1}{2}$$

$$D(p_{FB}^H, H) = 1 - p_{FB}^H \cdot \frac{1}{q} = 1 - \frac{H + c}{2} \cdot \frac{1}{H} = \frac{H - c}{2H}$$

Profits are then

$$\pi_{FB}^L = (p_{FB}^L - c_L)D(p_{FB}^L, L) = \left(\frac{1}{2}L - 0\right) \cdot \frac{1}{2} = \frac{1}{4}L$$

$$\pi_{FB}^H = (p_{FB}^H - c_H)D(p_{FB}^H, H) = \left(\frac{H + c}{2} - c\right) \cdot \frac{H - c}{2H} = \frac{(H - c)^2}{4H}$$

3

We should think of this as a signalling game: the monopolist sets a price in order to signal which product this is

Sender is the monopolist

Receivers are the consumers

4

The L-monopolist can deviate by pretending to sell an H product because:

$$(p_{FB}^H - c_L)D(p_{FB}^H, H) = \frac{H+c}{2} \cdot \frac{H-c}{2H} = \frac{H^2 - c^2}{4H} > \frac{1}{4}L = \pi_{FB}^L$$

So no, the FB equilibrium is not a possible SB equilibrium unless the single-crossing property is satisfied. The inequality is valid since

$$\frac{H^2 - c^2}{4H} > \frac{1}{4}L \Leftrightarrow H(H - L) > c^2$$

5

Profits were

$$\pi_H = (p - c) \left(1 - \frac{p}{q}\right)$$

$$\pi_L = p \left(1 - \frac{p}{q}\right)$$

Partial derivatives

$$\frac{\partial \pi_H}{\partial p} = 1 + \frac{c}{q} - \frac{2p}{q}$$

$$\frac{\partial \pi_L}{\partial p} = 1 - \frac{2p}{q}$$

And since

$$1 + \frac{c}{q} - \frac{2p}{q} > 1 - \frac{2p}{q} \Leftrightarrow \frac{c}{q} > 0$$

The single-crossing property is satisfied

6

Since in a separating equilibrium in SB, the L monopolist is easily identified because of the low quality, the price she should charge is the price such that π_{FB}^H is maximised so $p_{FB}^L = \frac{1}{4}L$ as we found before.